

Rechnen im Binärsystem

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$$\begin{array}{r}
 1) \quad \begin{array}{r} 1101 \\ + 0110 \\ \hline \end{array} \begin{array}{l} \hat{=} 13_{10} \\ \hat{=} 6_{10} \end{array} \\
 \text{Ü: } 1 \\
 \text{S: } \underline{10011} \hat{=} 19_{10}
 \end{array}$$

$$\begin{array}{r}
 2) \quad \begin{array}{r} 10010,00 \\ + 10,01 \\ \hline \end{array} \begin{array}{l} \hat{=} 18_{10} \\ \hat{=} 2,5_{10} \end{array} \\
 \text{Ü: } 1 \\
 \text{S: } \underline{10100,01} \hat{=} 20,5_{10}
 \end{array}$$

$$\begin{array}{r}
 3) \quad \begin{array}{r} 1,11101 \\ + 11,01101 \\ \hline \end{array} \\
 \text{Ü: } \\
 \text{S: } \underline{101,01010}
 \end{array}$$

$$\begin{array}{r}
 4) \quad \begin{array}{r} 111 \\ - 101 \\ \hline \end{array} \\
 \text{Ü: } \\
 \text{S: } \underline{010}
 \end{array}$$

$$\begin{array}{r}
 5) \quad \begin{array}{r} 1,1101 \\ - 1,1010 \\ \hline \end{array} \\
 \text{Ü: } 1 \\
 \text{S: } \underline{0,0011}
 \end{array}$$

$$\begin{array}{r}
 6) \quad \begin{array}{r} 11,00101 \\ - 0,11111 \\ \hline \end{array} \\
 \text{Ü: } 1 \quad 1 \\
 \text{S: } \underline{10,00110}
 \end{array}$$

$$\begin{array}{r}
 7) \quad \begin{array}{r} 11011 \cdot 1001 \\ 11011 \\ + 00000 \\ + 00000 \\ + 11011 \\ \hline \end{array} \\
 \text{Ü: } 1 \quad 1 \\
 \text{S: } \underline{11110011}
 \end{array}$$

$$\begin{array}{r}
 8) \quad \begin{array}{r} 1,11 \cdot 0,1 \\ 000 \\ + 111 \\ \hline \end{array} \\
 \text{S: } \underline{0,111}
 \end{array}$$

$$\begin{array}{r}
 9) \quad \begin{array}{r} 100,11101 \cdot 11,0111 \\ 10011101 \\ + 10011101 \\ + 00000000 \\ + 10011101 \\ + 10011101 \\ + 10011101 \\ \hline \end{array} \\
 \text{Ü: } 1 \quad 1 \quad 1 \\
 \text{S: } \underline{10000,110111011}
 \end{array}$$

$$\begin{array}{r}
 10) \quad 11,11 : 11 = \underline{101} \\
 - 11 \\
 \hline 001 \\
 - 00 \\
 \hline 011 \\
 - 11 \\
 \hline 000
 \end{array}$$

$$\begin{array}{r}
 11) \quad 10000 : 101 = 11, \underline{0011} \\
 - 101 \\
 \hline 0110 \\
 - 101 \\
 \hline 0010 \\
 - 000 \\
 \hline 00100 \\
 - 000 \\
 \hline 1000 \\
 - 1001 \\
 \hline 00110 \\
 - 101 \\
 \hline 000010
 \end{array}$$

$$\begin{array}{r}
 12) \quad 10010111,001 : 1011101 = \underline{\underline{1,101}} \\
 \underline{-11011101} \\
 001110100 \\
 \underline{-1011101} \\
 0000101110 \\
 \underline{-00000000} \\
 001011101 \\
 \underline{-1011101} \\
 00000000
 \end{array}$$